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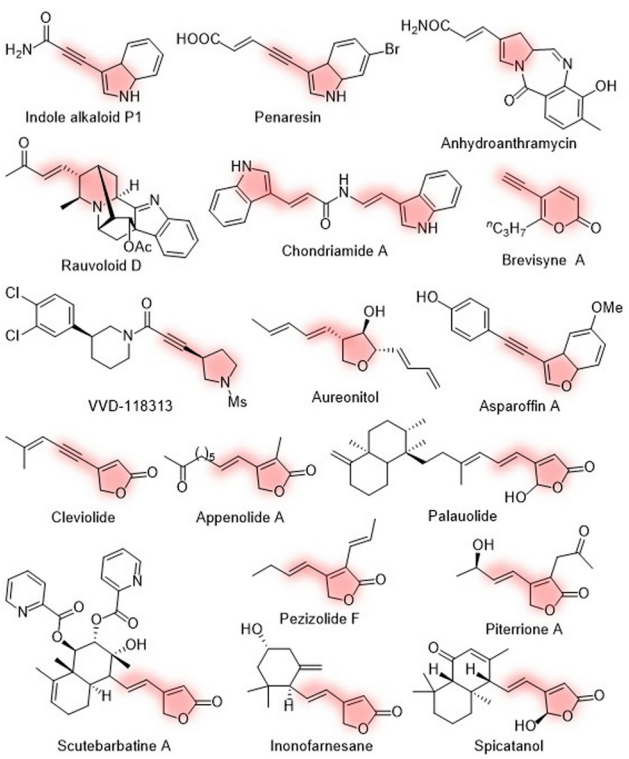
Aerobic bimetallic catalysis for oxy-alkynylation of allenes†

Junjie Fan, ‡ Dongjie Zhang, ‡ Yaqi Shi, Chunling Fu and Shengming Ma *

Here we report a conceptual protocol to construct 4-alkynyl furan-2(5*H*)-one derivatives efficiently through Rh(III)/Cu(I) co-catalyzed highly regioselective oxy-alkynylation of 2,3-allenoic acids with terminal alkynes under mild aerobic conditions with atom economy. A wide variety of functional groups including biologically active groups have been tolerated. The synthetic potentials have also been demonstrated and the total syntheses of natural product appenolide A and its (*Z*)-isomer have been achieved by applying this protocol as the key step efficiently. A mechanism has been proposed based on mechanistic studies.

Introduction

Alkynyl or (*Z*) or (*E*)-alkenyl-substituted cyclic structural units are commonly observed in natural and/or biologically active products. For example, aza- and oxa-cyclic structurals bearing alkynyl or alkenyl exist in indole alkaloid P1, penaresin, anhydroanthramycin, rauvoloid D, chondriamide A, brevisyne A, VVD-118313, aureonitol, asparoffin A, cleviolide, appenolide A, palauolide, Scutebarbatine A, pezizolide F, piterrione A, inonofarnesane, and spicatanol, *etc.* (Scheme 1).[1] As we know, Sonogashira coupling reactions of electrophilic organic halides with terminal alkynes have been established as one of the most efficient ways for the syntheses of substituted alkynes (Scheme 2a).[2,3] We envisioned a concept of the coupling reaction of cyclo alkenyl or alkyl metallic species *in situ* formed *via* the cyclic nucleometalation of unsaturated hydrocarbons with a nucleophilic functionality with terminal alkynes for the efficient synthesis of alkynyl-substituted cyclic skeleton (Scheme 2b), which should be a platform reaction for the selective synthesis of alkynyl or (*Z*) or (*E*)-alkenyl-substituted cyclic compounds listed in Scheme 1. The key here is the match of the reactivity of these two species, the regeneration of the catalyst, Glaser–Hay coupling reaction of the terminal alkynes applied, and the regioselectivity of the cyclization reaction.[4,5] Here we wish to report the realization of such a concept, *i.e.*, Rh(III)/Cu(I) co-catalyzed highly selective oxy-alkynylation of 2,3-allenoic acids[6] with terminal alkynes applying oxygen in air as the green oxidant for the efficient regeneration of the catalytically active tri-valent rhodium species (Scheme 2c).



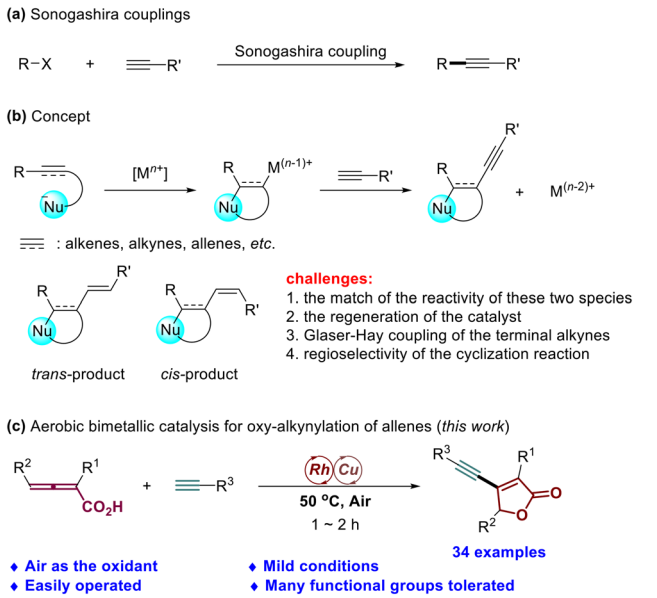
Scheme 1 Natural and/or biologically active products containing β-C–C triple or double bonds.

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Scheme 2 Transition metal-catalyzed coupling/cyclization of allenes with terminal alkynes.

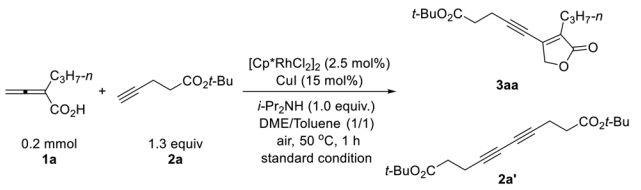
Results and discussion

Our efforts commenced with a model reaction of 2-propyl-2,3-butadienoic acid **1a** and *tert*-butyl pent-4-ynoate **2a**. After tedious optimization, the optimal reaction conditions have been defined: the reaction of allenoic acid **1a** (0.2 mmol) and terminal alkyne **2a** (1.3 equiv.) under the catalysis of $[Cp^*RhCl_2]_2$ (2.5 mol%) and CuI (15 mol%) with i-Pr_2NH (1.0 equiv.) as the base in a 1 : 1 mixture of DME (1,2-dimethoxyethane)/toluene (0.2 M) under air atmosphere at 50 °C giving 4-alkynylfuran-2(5*H*)-one **3aa** in 71% NMR yield (Table 1, entry 1). Meanwhile, the formation of Glaser–Hay coupling reaction product diyne **2a′** was also observed.[7] Some other metal catalysts, such as $PdCl_2$, $Pd(PPh_3)_2Cl_2$, $Pd(PhCN)_2Cl_2$, $Pd(OAc)_2$, $[Cp^*IrCl_2]_2$, Cp^*RuCl_2, $Ni(OAc)_2{\cdot}4H_2O$, and $Cp^*CoI_2(CO)$ were surveyed, with no better results (entries 2–9). Upon using the single solvent of toluene or DME, the yield was slightly lower (entries 10 and 11). Increasing or decreasing the loading of terminal alkyne **2a** failed to improve the yield (entries 12 and 13). Interestingly, the oxygen in air is necessary for this reaction, and using the pure oxygen resulted in a much lower yield (entries 14 and 15). Each of $[Cp^*RhCl_2]_2$, CuI, and i-Pr_2NH are required (entries 16–18).

With the optimal conditions in hand, the scope of oxy-alkynylation of 2,3-allenoic acids **1** and terminal alkynes **2** was then investigated (Table 2). The terminal alkynes with simple 1°-alkyl chains **2b–2d** reacted with 2,3-allenoic acid **1a** to afford corresponding products **3ab–3ad** in decent yields. The strained cyclopropyl-substituted acetylene also gave the desired product **3ae**. The alkynes bearing highly reactive functional groups, such as *tert*-butoxycarbonyl group (**2a**), halide (Cl and Br), hydroxy and protected hydroxy, all could be tolerated giving the corresponding 2-furan-5(*H*)-ones **3aa** and **3af–3ak** in 60%–70% yields. Cyano and thiocyanato with remarkable coordinating ability could also be tolerated to afford **3al** and **3am** in excellent yields. Besides the alkyl alkynes, different aryl-substituted acetylenes could also afford target products **3an** and **3ao** smoothly. Menthol (**2p**), steroid skeleton (**2q** and **2r**), and (*S*)-ibuprofen (**2s**) may all be incorporated to afford the related products **3ap**~(*S*)-**3as** in excellent yields.

Then differently substituted 2,3-allenoic acids were further investigated with some terminal alkynes: the R^1 of allenoic acids may be methyl, phenylethyl, chloropropyl, and alkoxyalkyl groups *etc.* producing **3ba–3ea**. In addition, the C–C double and triple bonds remained untouched affording **3fa** and **3ga**. Synthetically versatile functionalities such as F, Cl, Br, OMe, and CF_3 may be installed on the benzyl group affording **3ha–3ma** in moderate yields. The structure of the products has further been confirmed by the X-ray diffraction study on the single crystal of **3ma**. With trisubstituted allenoic acids, the yields are lower, indicating an obvious steric effect (**3nh–3ph**). The *in situ* formed 5-phenylfuran-2(5*H*)-one from the reaction of **1p** with **2b** was further oxidized affording γ-hydroxylated product **5** by air.[8]

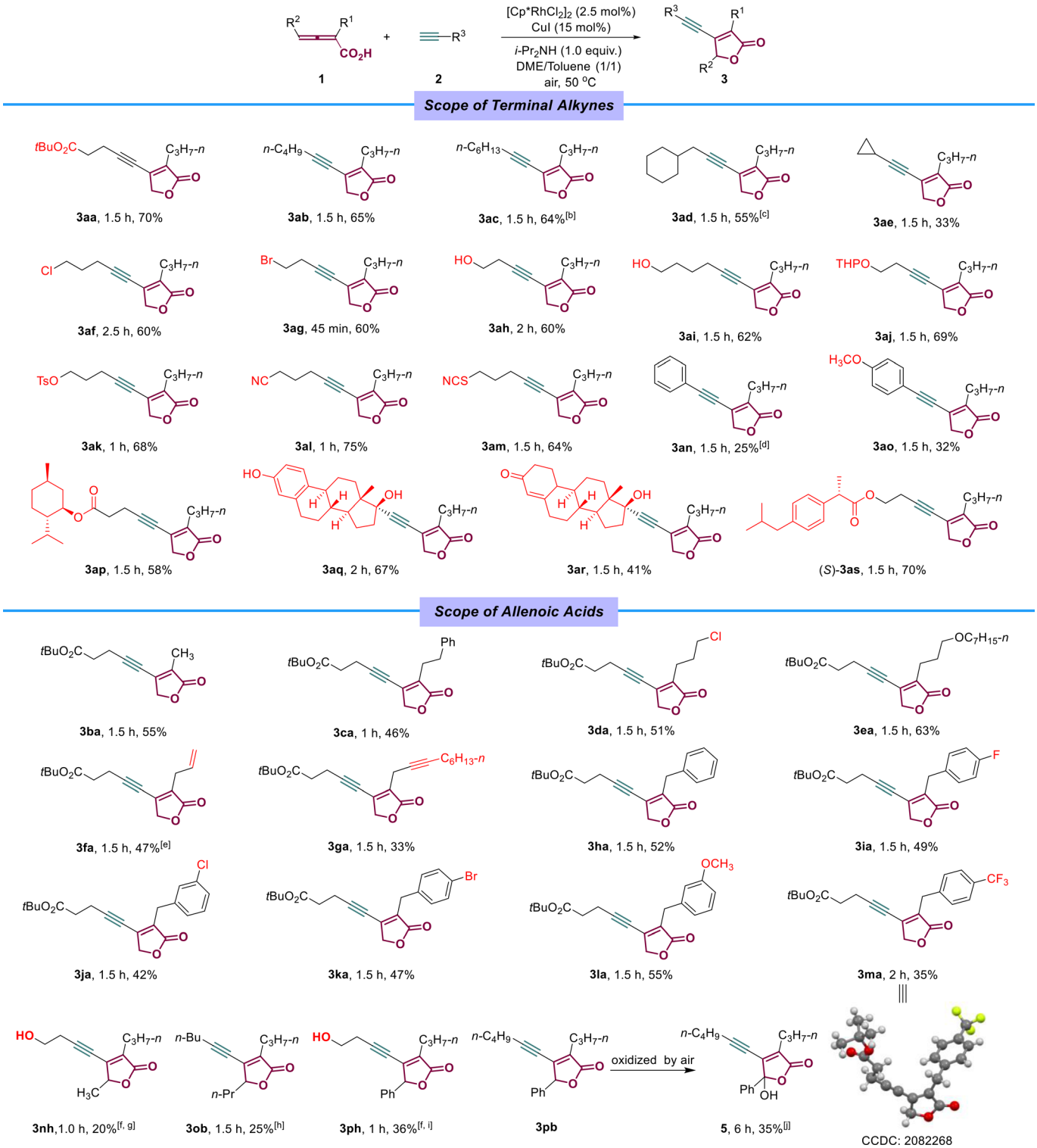
Table 1 Optimization of the reaction conditions[a]



Entry	Variations from standard conditions	NMR yield of **3aa**/**2a′** [b] (%)
1	None	71/13[c]
2	5 mol% $PdCl_2$ instead of $[Cp^*RhCl_2]_2$	11/62
3	5 mol% $Pd(PPh_3)_2Cl_2$ instead of $[Cp^*RhCl_2]_2$	13/—
4	5 mol% $Pd(PhCN)_2Cl_2$ instead of $[Cp^*RhCl_2]_2$	11/28
5	5 mol% $Pd(OAc)_2$ instead of $[Cp^*RhCl_2]_2$	8/—
6[d]	2.5 mol% $[Cp^*IrCl_2]_2$ instead of $[Cp^*RhCl_2]_2$	1/—
7[e]	5 mol% Cp^*RuCl_2 instead of $[Cp^*RhCl_2]_2$	1/—
8[f,g]	5 mol% $Ni(OAc)_2{\cdot}4H_2O$ instead of $[Cp^*RhCl_2]_2$	2/29
9[f]	5 mol% $Cp^*CoI_2(CO)$ instead of $[Cp^*RhCl_2]_2$	3/48
10	Toluene instead of toluene/DME/(1/1)	54/—
11	DME instead of toluene/DME (1/1)	69/—
12	1.2 equiv. of **2a**	57/—
13	1.4 equiv. of **2a**	66/—
14	O_2 instead of air	50/—
15[h]	N_2 instead of air	5/—
16[i]	Without $[Cp^*RhCl_2]_2$	2/—
17[j]	Without CuI	0/—
18	Without i-Pr_2NH	8/—

[a] The standard reactions were conducted with **1a** (0.2 mmol), **2a** (0.26 mmol), $[Cp^*RhCl_2]_2$ (2.5 mol%), CuI (15 mol%), and i-Pr_2NH (1.0 mmol) in DME (0.5 mL)/toluene (0.5 mL) at 50 °C under air atmosphere for 1 h in a 25 mL Schlenk vessel. [b] Yields of **3aa** and **2a′** were determined by ^1H NMR using mesitylene as the internal standard, the yield of **2a′** determined was not precise due to lack of characteristic peaks. [c] Isolated yield of **2a′**. [d] 6% of 4-iodo-3-propylfuran-2(5*H*)-one **4** was determined. [e] 3% of 4-iodo-3-propylfuran-2(5*H*)-one **4** was determined. [f] 12 h. [g] 9% of **1a** was recovered. [h] 82% of **2a** was recovered. [i] 60% of **2a** was recovered. [j] 1.5 h, 62% of **2a** was recovered.

Table 2 Reaction scope[a]

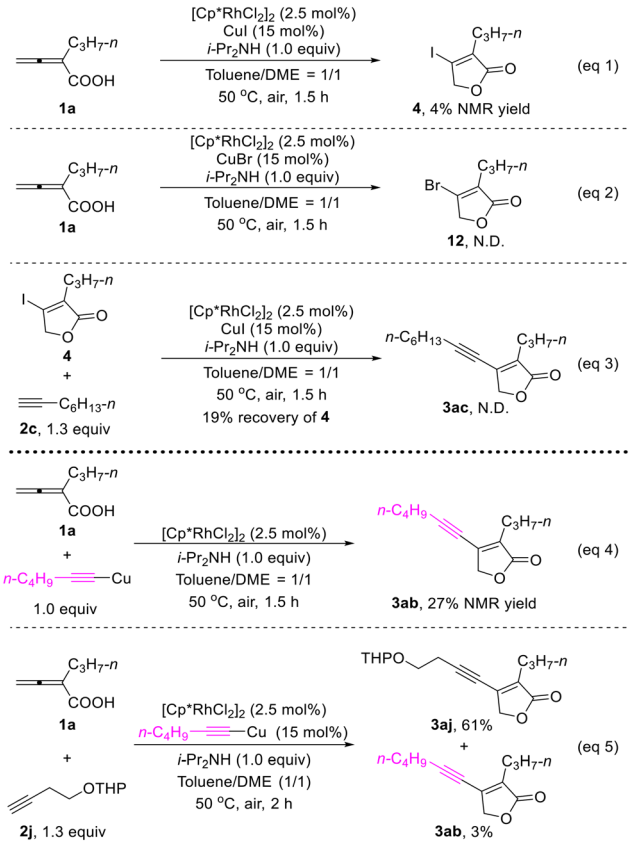


[a] The reaction was conducted with **1** (1.0 mmol), **2** (1.3 mmol), $[Cp^*RhCl_2]_2$ (2.5 mol%), CuI (15 mol%), and i-Pr_2NH (1.0 mmol) in DME (2.5 mL)/toluene (2.5 mL) at 50 °C under air atmosphere in a 100 mL Schlenk vessel unless otherwise noted. [b] The reactions were conducted in DME (0.65 mL) and toluene (0.65 mL). [c] A mixture of **3ad** and 2% of 4-iodo-3-propylfuran-2(5*H*)-one **4** was isolated. [d] 22% of diacetylene product **2n′** was isolated. [e] 20% of diacetylene product **2a′** was isolated. [f] 0.5 mmol scale, $[Cp^*RhCl_2]_2$ (5 mol%), CuI (20 mol%), i-Pr_2NH (1.0 mmol) in DME (3.5 mL)/dioxane (3.5 mL). [g] 57% of 5-methyl-3-propylfuran-2(5*H*)-one **1n′** was determined by ^1H NMR. [h] 5 mL of CH_3CN were used, 68% of 3,5-dipropylfuran-2(5*H*)-one **1o′** was isolated. [i] 5% of 5-phenyl-3-propylfuran-2(5*H*)-one **1p′** was determined by ^1H NMR. [j] 0.2 mmol scale in toluene (1.0 mL) in a 25 mL Schlenk vessel, 6 h.

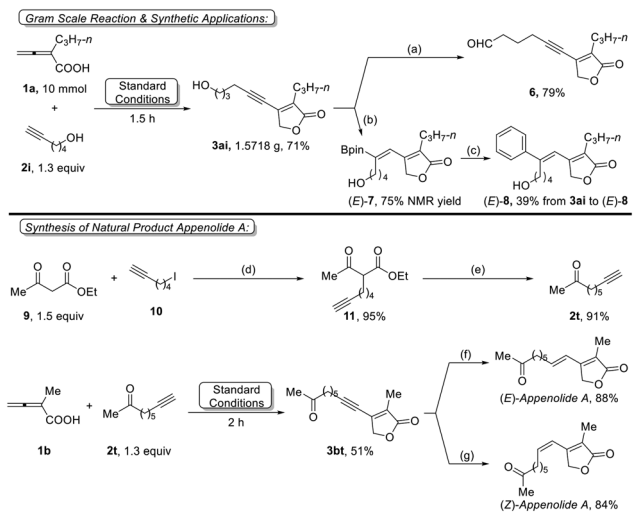
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In order to explore the synthetic potentials of this protocol, a gram-scale reaction has been conducted under the standard conditions affording **3ai** in 71% yield. The hydroxy group in **3ai** could be oxidized in dichloroethane to produce the aldehyde **6** in air by using $Fe(NO_3)_3{\cdot}9H_2O$, 4-OH-TEMPO, and NaCl as the catalysts.[9] CuCl-catalyzed hydroboration[10] of **3ai** produced an alkenyl borate (*E*)-**7** in 75% NMR yield, which underwent cross-coupling with iodobenzene catalyzed by $Pd(PPh_3)_4$ to yield product (*E*)-**8** in 52% yield (39% from **3ai** to (*E*)-**8**).[11] Appenolide A, isolated from liquid cultures of the coprophilous fungus *Podospora appendiculata*, exhibited both antibacterial and antifungal activities.[1m] Alkylation of ethyl acetoacetate **9** with 6-iodo-1-hexyne **10** afforded terminal alkyne **11** in 95% yield. Subsequent decarboxylation afforded terminal alkynyl methyl ketone **2t** in 91% yield. Its reaction with 2-methyl-2,3-butadienoic acid **1b** under the standard conditions afforded **3bt** in 51% yield. $Cp^*Ru(CH_3CN)_2PF_6$-catalyzed *trans*-hydrosilylation[12] and the desilylation[13] afforded appenolide A in 88% combined yield for the last two steps while (*Z*)-isomer of appenolide A was provided through *cis*-hydrogenation catalyzed by Lindlar catalyst in 84% yield (Scheme 3).

To clarify the mechanism, a series of control reactions were conducted: when allenoic acid **1a** was stirred under the standard conditions in the absence of a terminal alkyne, 4% NMR yield of β-iodofuran-2(5*H*)-one **4** was detected (Scheme 4, eqn (1)). However, using CuBr instead of CuI, the corresponding bromide **12** was not formed (eqn (2)). Using **4** instead



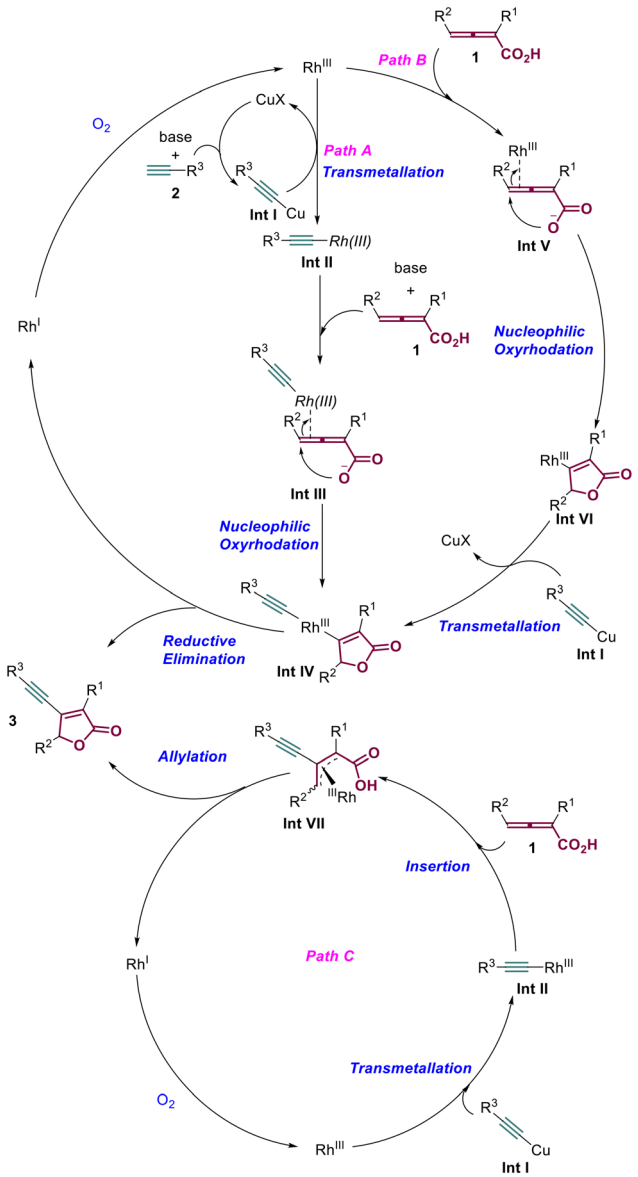
Scheme 4 Mechanistic studies.



Scheme 3 Gram scale reaction & synthetic applications, synthesis of natural product appenolide A. Reaction conditions: (a) $Fe(NO_3)_3{\cdot}H_2O$ (10 mol%), 4-OH-TEMPO (10 mol%), NaCl (10 mol%), DCE, air bag (1 atm), 25 °C, 23.5 h. (b) $B_2(Pin)_2$ (2.5 equiv.), CuCl (5 mol%), PPh_3 (5 mol%), KO*t*-Bu (50 mol%), MeOH (4.0 equiv.), THF, 25 °C, 20 h. (c) PhI (1.2 equiv.), $Pd(PPh_3)_4$ (10 mol%), Cs_2CO_3 (3.0 equiv.), THF, 60 °C, 14.5 h. (d) NaH (1.5 equiv.), THF, 0 °C, 10 min, then 80 °C, 12 h. (e) NaOH (4.0 equiv.), MeOH/H_2O = 2/1, 100 °C, 2.5 h. (f) step 1: $(EtO)_3SiH$ (2.0 equiv.), $Cp^*Ru(CH_3CN)_2PF_6$ (5 mol%), DCM, 0 °C, then rt, 6 h. step 2: AgF (2.0 equiv.), MeOH, rt, 12 h. (g): Lindlar Pd (2.5 mol%), quinoline (1.25 equiv.), ethyl acetate, H_2 bag (1 atm), rt, 24 h.

of 2,3- allenoic acid **1a** failed to afford the target product **3ac** under the standard conditions (eqn (3)), suggesting that **4** was not the intermediate for this transformation.[14] When 1.0 equiv. of copper(I) butylacetylenide was added instead of terminal alkynes, target product **3ab** was formed in 27% NMR yield (eqn (4)). In addition, when 15 mol% of copper(I) butylacetylenide was applied instead of CuI in the presence of terminal alkyne containing OTHP group **2j**, the reaction afforded **3aj** in 61% yield with the product **3ab** with the alkynyl from copper(I) butylacetylenide in 3% yield (eqn (5)), demonstrating that alkynyl copper was the intermediate.

On the basis of these experimental data, a mechanism was proposed as shown in Scheme 5. In path A, the terminal alkyne would be converted to the alkynyl copper **Int I** in the presence of base and copper(I). After transmetallation between **Int I** and the rhodium(III) catalyst, alkynyl rhodium(III) **Int II**, which would trigger the nucleorhodation[15] of 2,3-allenoic acid **1** to afford **Int IV** *via* the coordination complex **Int III**. Reductive elimination would afford the product **3** and rhodium(I) species, which would be oxidized back to the catalytically active rhodium(III) species to furnish the catalytic cycle with the oxygen from air as the green oxidant.[16] Alternatively, the nucleorhodation may take place firstly to afford furan-2(5*H*)-one-4-yl rhodium intermediate **Int VI**. Transmetallation between **Int VI** and **Int I** would also afford the same intermediate **Int IV** (Path B). There is still an alternative pathway, which



Scheme 5 Proposed mechanism.

could not be excluded (Path C):[17] the 2,3-allenoic acid **1** may also insert into the C–Rh bond of **Int II** to afford π-allyl Rh(III) species **Int VII**. Then the allylation of **Int VII** would afford the product **3** and rhodium(I) species.

Conclusions

In conclusion, we have developed an efficient Rh(III)/Cu(I) co-catalyzed regioselectivity-defined cyclic oxy-alkynylation of 2,3-allenoic acids with terminal alkynes, affording 4-alkynylfuran-2(5*H*)-one derivatives. The synthetic potentials have also been demonstrated. Both antifungal and antibacterial natural product, appenolide A and its *Z*-isomer have been synthesized highly stereoselectively by using this reaction as the key step. A Rh(III)/Rh(I) catalytic cycle has been proposed for the reaction.

Author contributions

S. Ma, J. Fan and D. Zhang designed the experiments. J. Fan, D. Zhang and Y. Shi performed the experiments. J. Fan and S. Ma wrote the manuscript. J. Fan, C. Fu and S. Ma checked the experimental data.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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